TrustSphere AI — Digital Evidence Verification Report

Generated 2026-10-23 16:43:03 UTC · Case ID ee318bfa-7acc-4259-a8c3-711b44473afc

1. Evidence Identification

Original filename	1_receipt_AUTHENTIC.png
File type	image
SHA-256 hash	`b4bb8adcfafebd4ddbddd89c14aa5c09bfa6cdbd2b63adf57277dd8bdaaaa07d`
Uploaded (UTC)	2026-10-23T16:43:03.562013
Analysis status	done

2. Forensic Verdict

INCONCLUSIVE — overall authenticity score 0.46 (0 = fake, 1 = authentic)

Manipulation type	GAN / AI-Generated Image (confidence 0.57)
Generator attribution	GAN family (e.g. StyleGAN) (confidence 0.40)

Reasoning:
• Overall authenticity score 0.46 → Inconclusive.
• Deep classifier flagged synthetic imagery (p=0.57).
• Error-level analysis shows inconsistent compression regions (splice).

3. Agent Findings

metadata — authenticity 0.450

Check	Status	Note
camera_info	suspicious	No camera make/model — common in AI-generated or re-encoded images
software	ok	No editing software tag
timestamps	suspicious	No EXIF timestamps present

image_forensics — authenticity 0.466

Check	Status	Note
ela	suspicious	Mean error level 31.67 (heatmap saved)
fft	ok	High-frequency energy ratio 0.325
deep_model	suspicious	Deepfake classifier authenticity 0.431
xai_gradcam	ok	Model-attention heatmap saved (gradcam)

security — authenticity n/a (no signal)

Check	Status	Note
c2pa_manifest	ok	No C2PA content credentials embedded (typical for most media).
invisible_watermark	skipped	Image too small for DWT-DCT decode.

4. Explainability Exhibits (XAI)

Exhibit — metadata feature importance (exact Shapley contributions; green → authentic, red → fake)

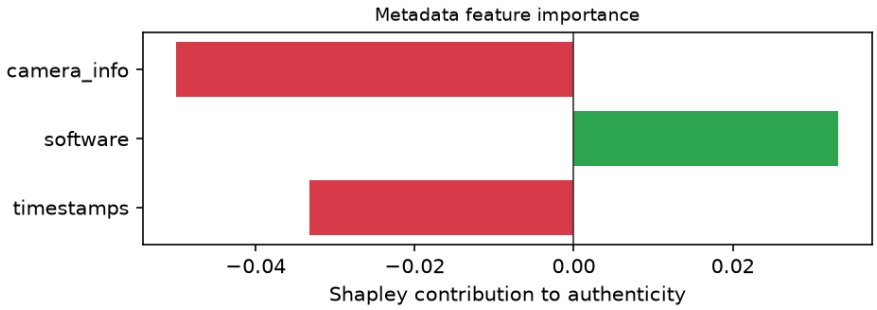


Exhibit — image_forensics: ELA heatmap (bright regions compress inconsistently → possible splice)

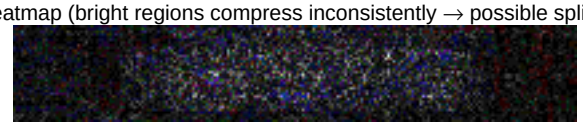
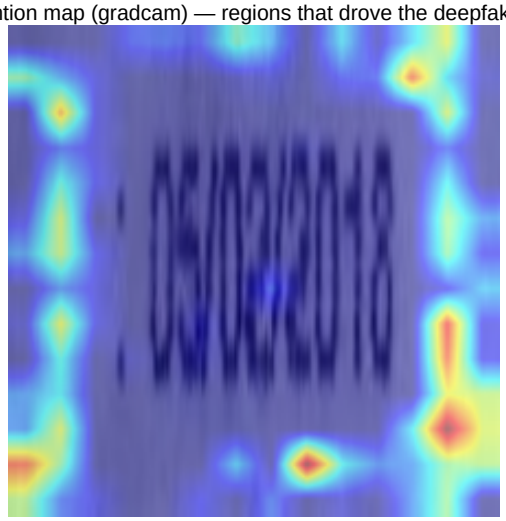


Exhibit — image_forensics: model attention map (gradcam) — regions that drove the deepfake classifier's decision



5. Chain of Custody

Event	Timestamp (UTC)	Detail
uploaded	2026-10-23T16:43:03.562013	
analyzed	2026-10-23T16:43:03.667950+00:00	
signed	2026-10-23T16:43:03.668087+00:00	`6217864d3e24af8f2157b4800b3b29c95f497c141297c36aae97212c7b77f184`
blockchain_skipped	2026-10-23T16:43:03.668101+00:00	`POLYGON_PRIVATE_KEY not set — anchoring disabled`

Blockchain network	polygon-amoy
Anchor status	skipped

6. Record Integrity — Digital Signature

The analysis record below is signed with the TrustSphere server's Ed25519 key. Any alteration of the signed payload invalidates the signature (verifiable via GET /api/verify/{file_id}).

Algorithm	Ed25519
Key fingerprint	`6217864d3e24af8f2157b4800b3b29c95f497c141297c36aae97212c7b77f184`
Signature	`1e23a8268dc77551949472cbafd019b074db7fad6d08b1555cb1900d57447028fca7acdf3eca288c8e4ba918d6f347a1a5937f8e6306e7a5f626af4958809e07`

This report was generated automatically by TrustSphere AI. The verdict is a statistical forensic assessment, not a legal determination.